

Introduction

Inverters are critical
Doing more than ever
Failing too often
Why they fail
What drives it
The opportunity

Backbone of grid connected PV, wind, battery systems
Ancillary services: reactive power, voltage regulation etc
43 to 70% of PV service requests, 0.48 failures/turbine year
Thermo mechanical fatigue from junction temperature cycling
High loading, reactive power, ambient temperature swings
Predict damage, anticipate failure, plan maintenance

Possible Solution

A tool that translates real inverter mission profiles into IGBT/diode power module life-time, enabling reliability-aware design, component selection and predictive maintenance.

Identified Research Gap

Table 1: No existing tool category spans the full mission profile to lifetime chain.

Tool	Mission profile	Long profiles	Unified chain	Open source	Device flexible	Reproducible	Electrical losses	Thermal model	Lifetime
FEM tools	△	×	△	×	✓	×	×	✓	×
Manufacturer tools	✓	△	✓	×	×	×	✓	✓	✓
Commercial sim.	✓	✓	△	×	✓	△	✓	✓	△
✓ Full	supported and integrated	△ Partial	possible but limited or indirect	×	None	not supported			

Research Contribution

PEARL (Power Electronics Age and Reliability)

An open source framework linking mission profiles, operating conditions and component specs to module reliability, through one transparent and reproducible pipeline.

Table 2: PEARL links the full mission profile to lifetime chain in one pipeline.

Tool	Mission profile	Long profiles	Unified chain	Open source	Device flexible	Reproducible	Electrical losses	Thermal model	Lifetime
PEARL	✓	✓	✓	✓	✓	✓	✓	✓	✓

Methodology

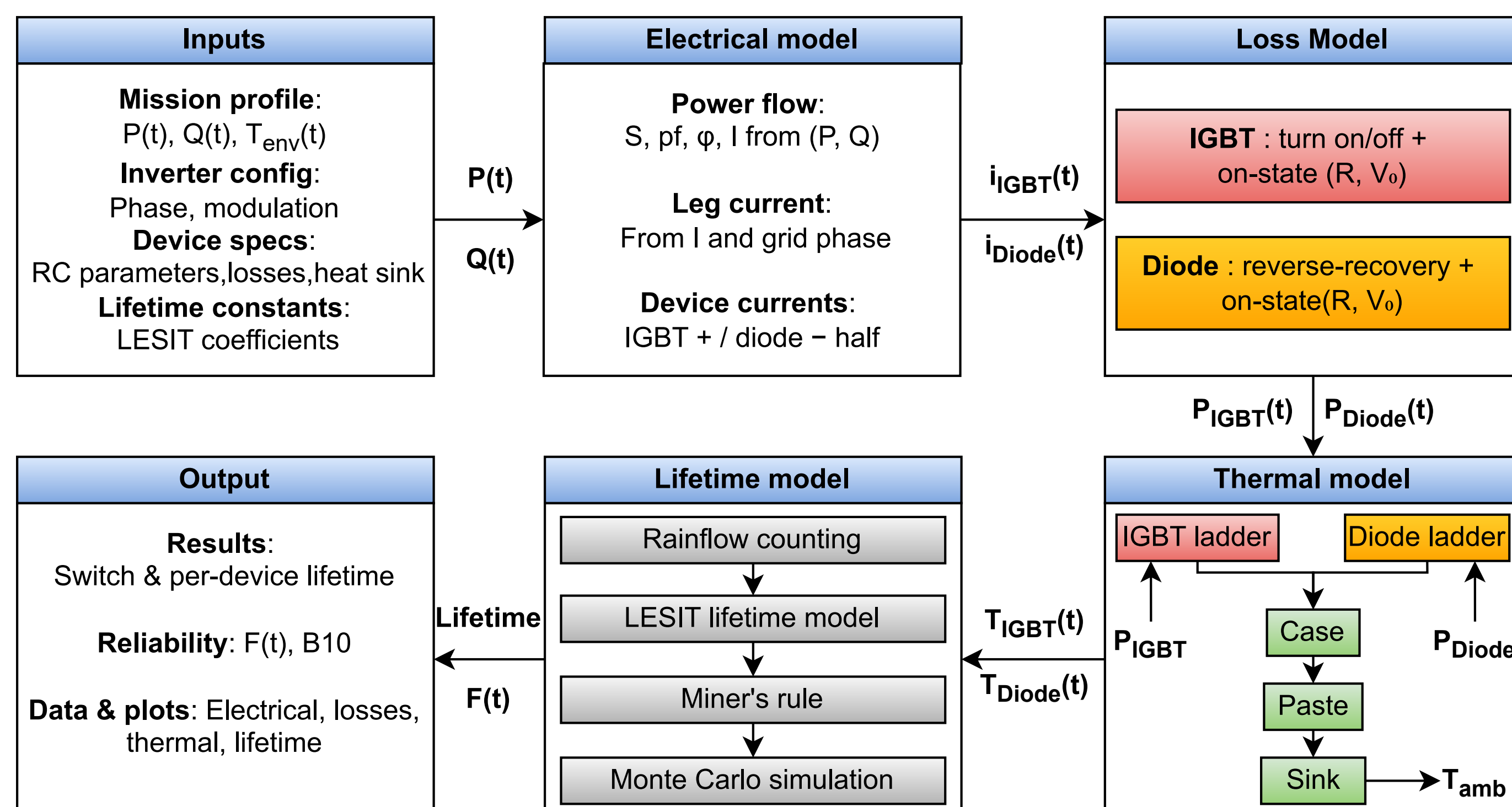


Figure 1: PEARL methodological framework for IGBT-diode lifetime assessment.

How does reactive power impact power module's lifetime?

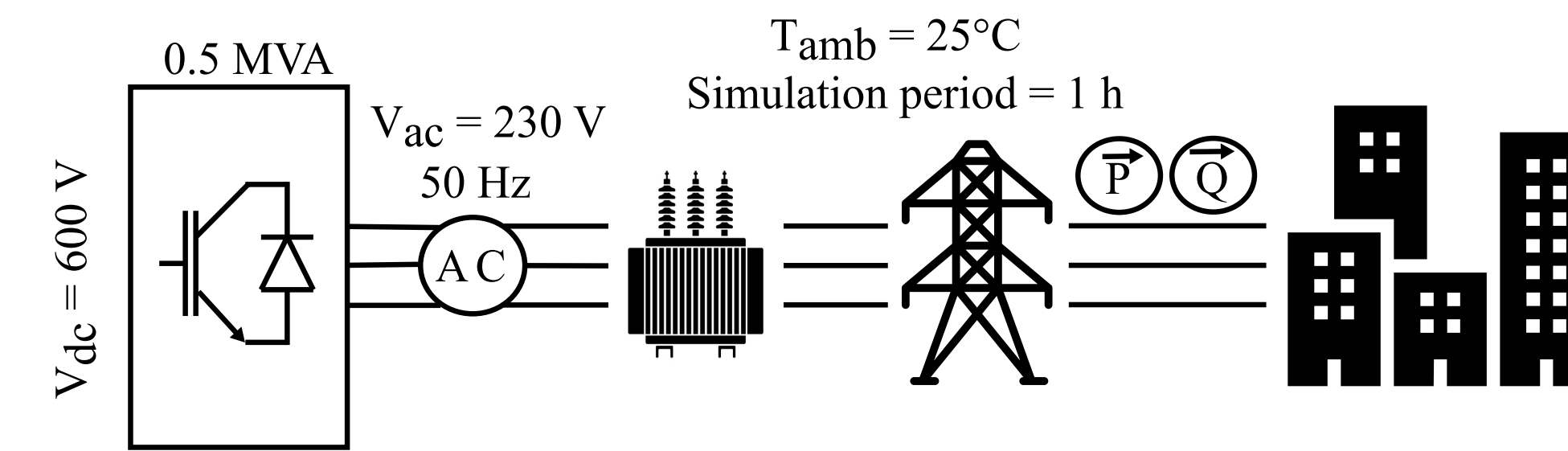


Figure 2: Grid connected inverter setup for the reactive power study.

Setup Infineon IKW50N60H3 (600 V, 50 A); 3-phase 0.5 MVA inverter, 230 V AC / 600 V DC, 50 Hz
Cooling Kryonaut paste and Cooling Innovations 3-121211U heat sink
Simulation Power factor swept 0 to 1 (ind & cap) at fixed apparent power over a 1 h cyclic profile

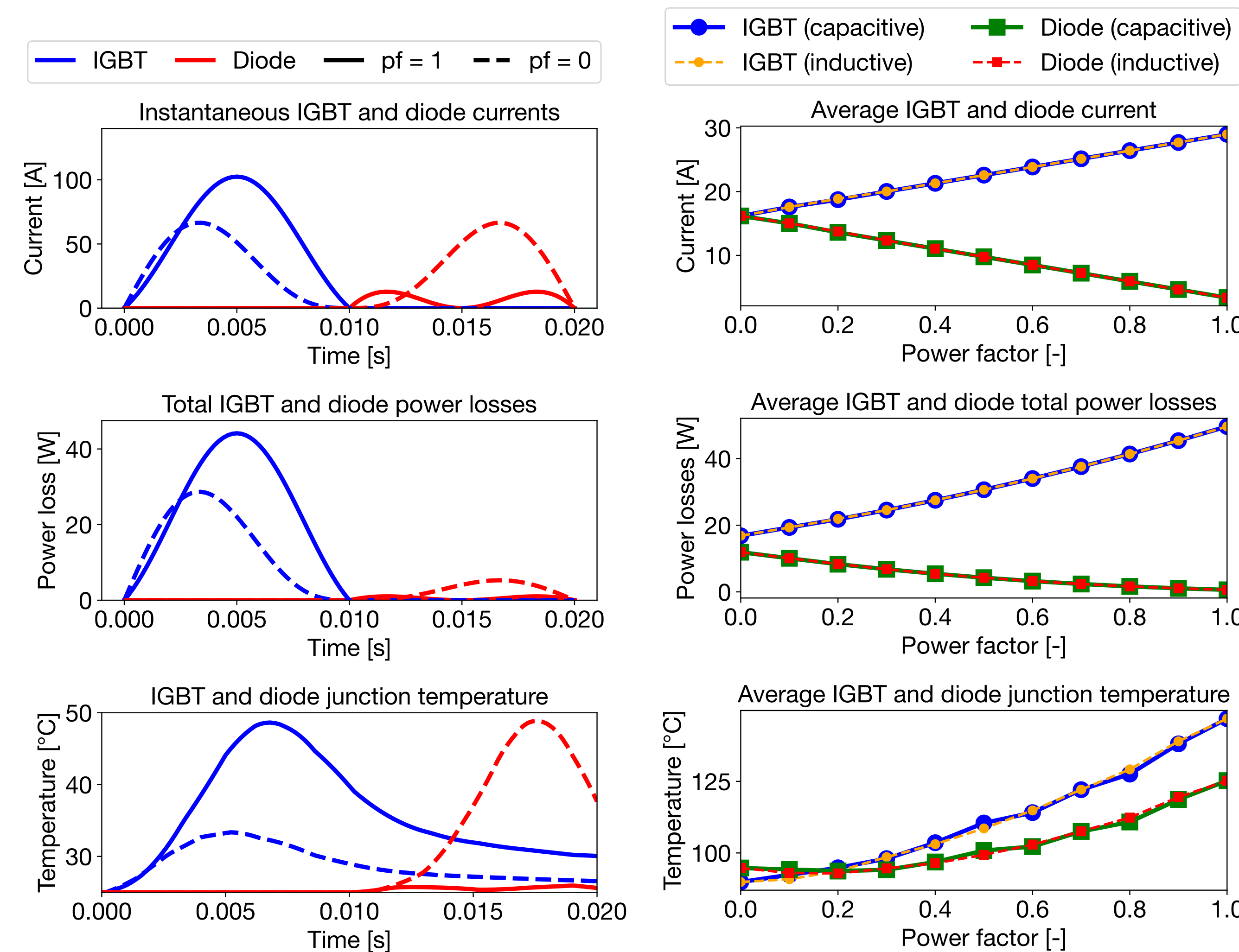


Figure 3: IGBT and Diode electro-thermal response.

Table 3: Impact of pf (1 to 0) on the IGBT and diode.

Quantity	IGBT	Diode
Figure 3		
Instantaneous current	falls	rises
Instantaneous losses	falls	rises
Temperature swing	falls	rises
Figure 4		
Average current	falls	rises
Total power losses	falls	rises
Junction temperature*	falls	falls
Estimated lifetime	rises	falls
*IGBT dominates case heating; as its losses fall, the diode cools despite rising losses. Inductive and capacitive give near identical results; magnitude matters, not sign.		

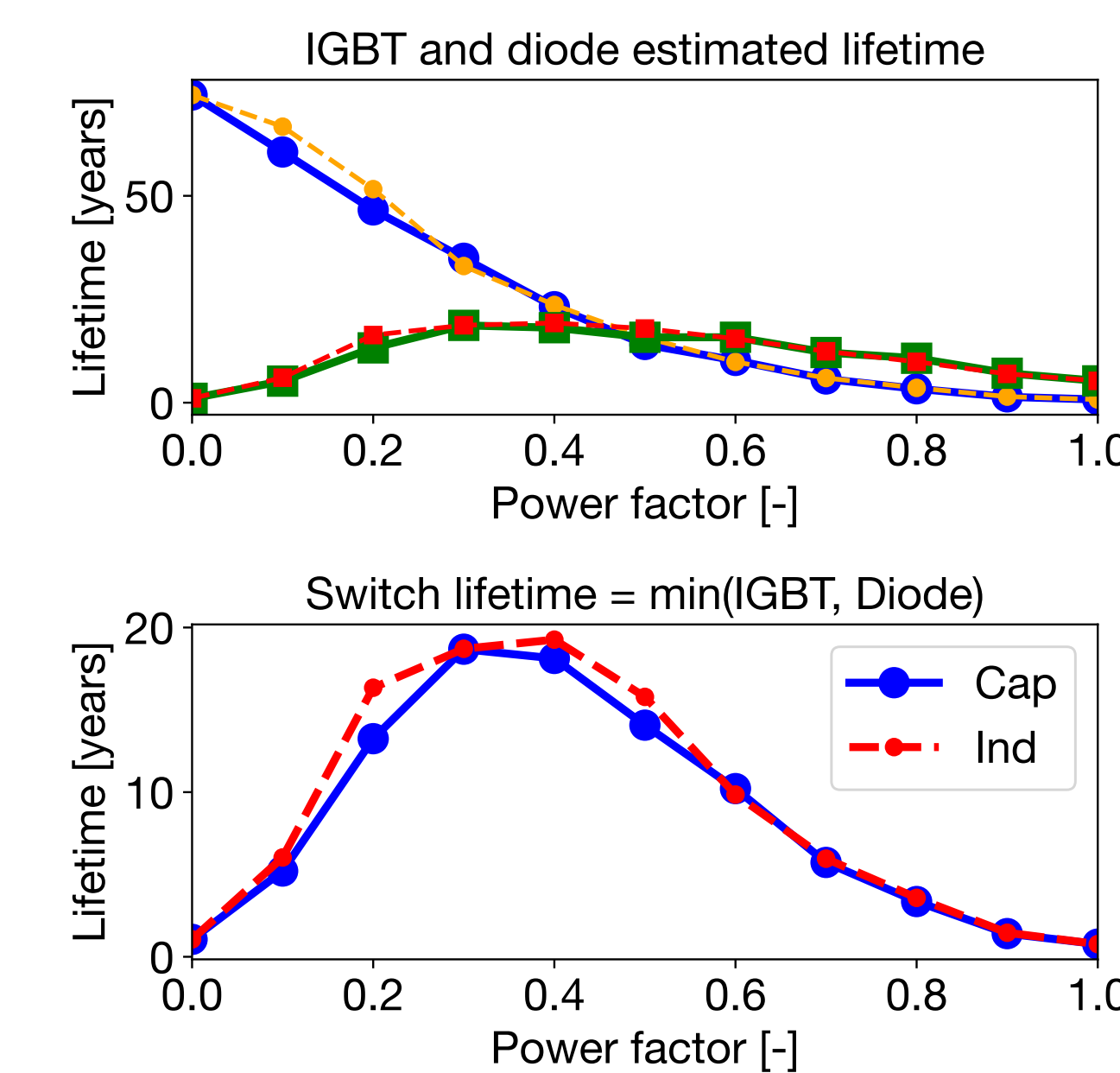


Figure 4: Impact of reactive power on IGBT and Diode current, losses, temperature and lifetime.

How does heat sink design impact module's reliability?

Setup Same as before, but power factor fixed at pf = 1
Constraint Junction temperature must stay below 175°C (datasheet limit)
Heat sinks Cooling Innovations 3-1212 (02U to 10U): higher number = larger = lower R_{th}
Air speed 1–4 m/s further lowers thermal resistance R_{th}

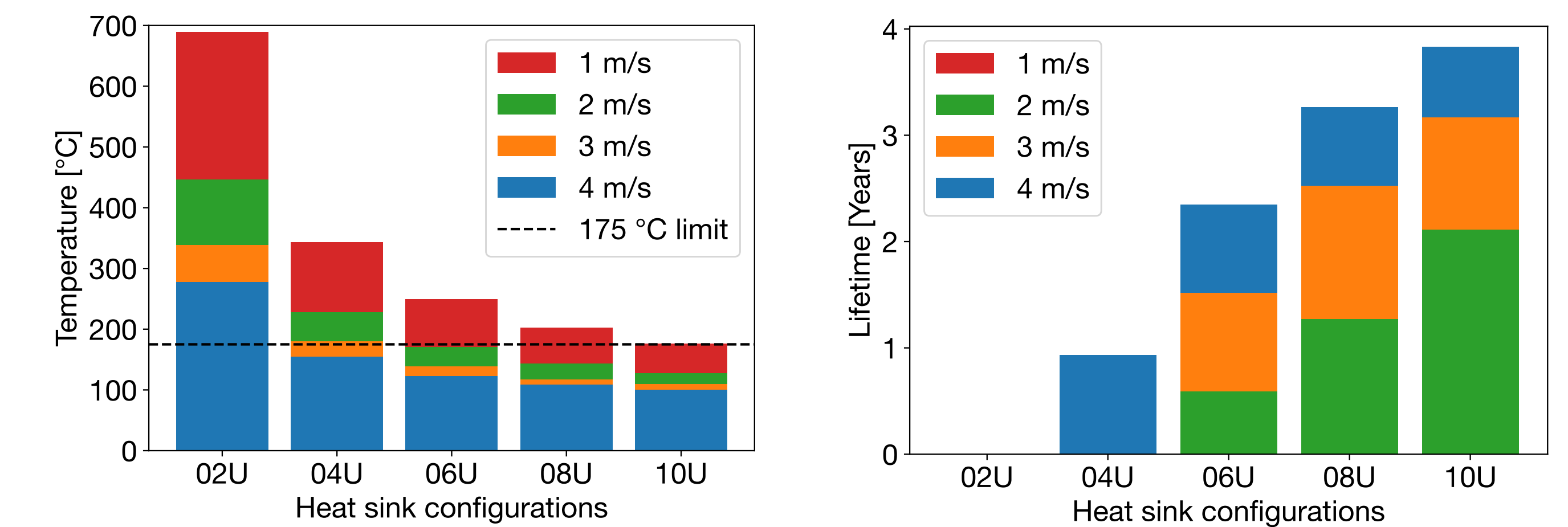


Figure 5: Maximum IGBT junction temperature per heat sink across air speeds (1–4 m/s).

Figure 6: Estimated module lifetime per heat sink across air speeds.

Table 4: Effect of larger sink and higher air speed.

Quantity	Sink size / Air speed increase
Heat dissipation	improves
Thermal resistance R_{th}	falls
Junction temperature	falls
Estimated lifetime	rises
B10 point (unreliability)	delayed
10U at 4 m/s is coolest and lives longest. - Larger sinks delay the B10 point. - Trade off: Compact heat sinks save space but need stronger forced cooling.	

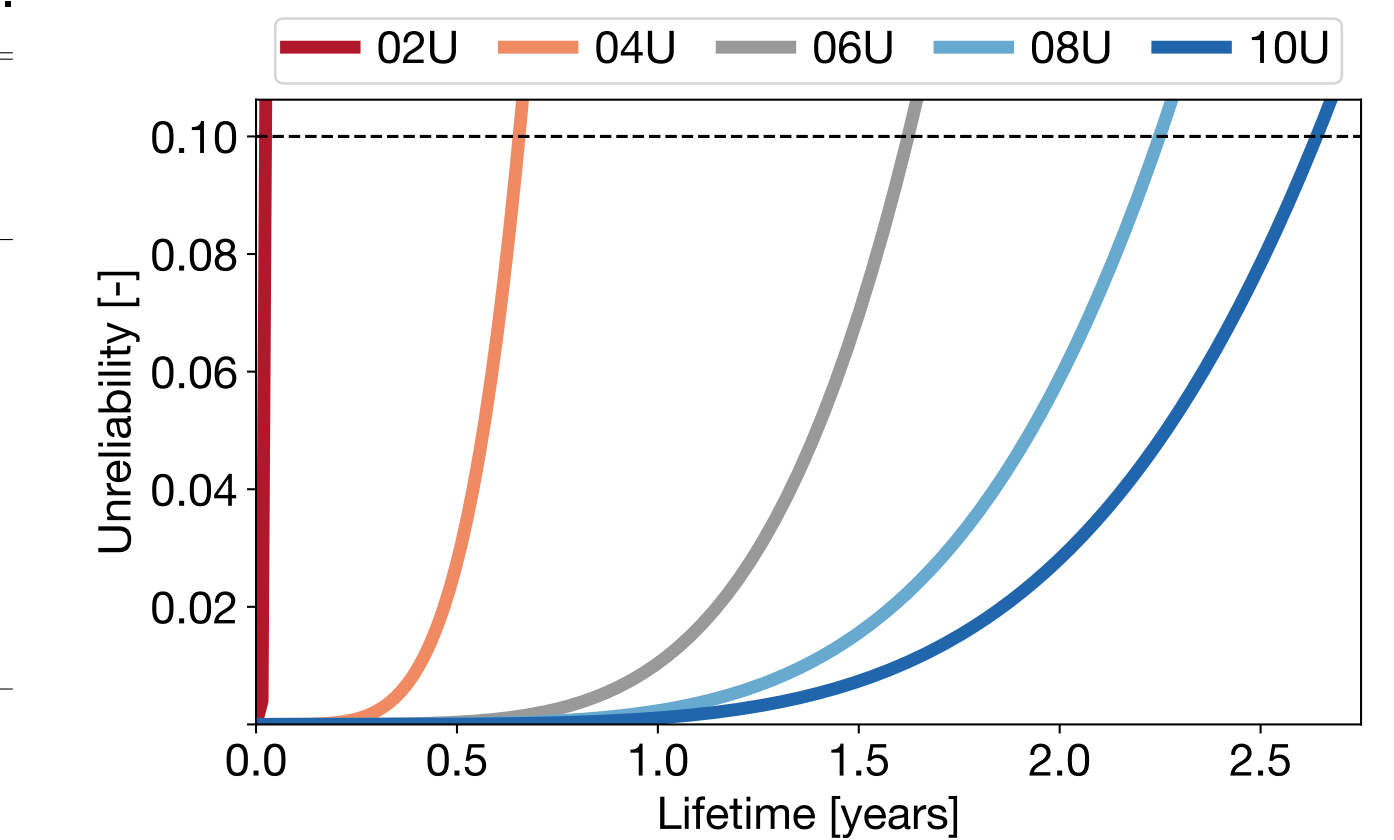


Figure 7: Switch unreliability $F(t)$ at 4 m/s per heat sink. Dashed line marks the B10 (10% failure) point.

Future Outlook

- Extend to other converter components critical for grid: DC link capacitors, LCL filters, MOSFETs etc.
- Build on component level reliability to model system level reliability of grid connected converters.

Goal

Bring reliability to the forefront of grid planning and decision making, linking component reliability to system level and grid level resilience.

PEARL Repository

Open & free
Data in, life out
Fast & flexible
Test before you build
Full reliability
Built to scale

Device physics to lifetime, fully transparent
No FEM or proprietary tools
Long profiles, any design
Any device, heat sink or cooling
Monte Carlo: distributions, B10
Toward system and grid level



Figure 8: PEARL repository.